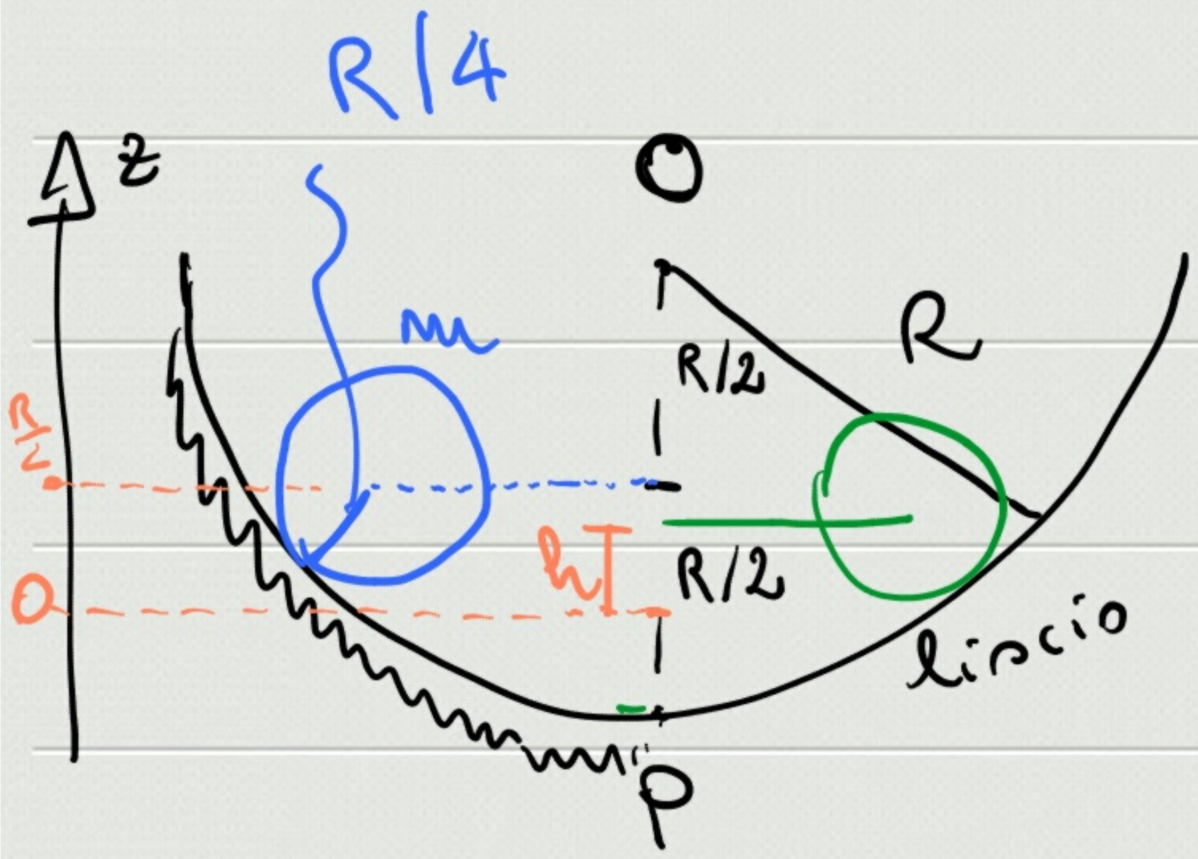

$v_{CM,0} = \omega_0 = 0$

Rotola senza strisciare

$h, \omega_{fin} = ?$

$$E_{m,i} = E_{m,f} \Rightarrow E_{p,in} = E_{k,fin}$$

$$mg\frac{R}{4} = \frac{1}{2} m v_{CM}^2 + \frac{1}{2} I_{CM}\omega^2 \qquad \left(v_{CM} = \omega\frac{R}{4}\right)$$

$$\Rightarrow \cancel{m}g\frac{\cancel{R}}{\cancel{4}} = \frac{1}{\cancel{2}}\cancel{m}\omega^2\frac{R^{\cancel{2}}}{\underset{8}{\cancel{16}}} + \frac{1}{\cancel{2}}\left(\frac{1}{\cancel{2}}\cancel{m}\frac{R^{\cancel{2}}}{16}\right)\omega^2$$

$$\Rightarrow g = \omega^2 R\left(\frac{1}{8} + \frac{1}{16}\right) \Rightarrow \boxed{\omega = 4\sqrt{\frac{g}{3R}}}$$

ω_{fin}

$$E_m = \text{cost}$$

$$\underbrace{\frac{1}{2} m v_{cm}^2 + \cancel{\frac{1}{2} I_{cm} \omega^2}}_{h_{in}=0} = \cancel{\frac{1}{2} I_{cm} \omega^2} + mgh$$

$$\frac{1}{2} \cancel{m} \omega^2 \frac{R^2}{16} = \cancel{m} g h$$

$$\Rightarrow h = \frac{\omega^2 R^2}{32 g} = \frac{\cancel{16} g}{3 \cancel{R}} \frac{R^{\cancel{2}}}{\frac{\cancel{32} g}{2}} = \frac{R}{6}$$